

ALP

WORKSHEET
TEACHING ACTIVITY

Algorithmics and Data Structures

COURSE UNIT

worksheet 09 – NumPy Library

In this exercise we will work with a **films.csv** dataset. This dataset stores data about films characterization and classification (made available for free, like many other datasets, through the <https://www.kaggle.com/datasets/platform>).

The dataset has the following attributes, illustrated in the figure.



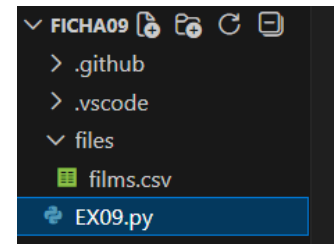
Typically, almost all available datasets require data cleaning. That is, they often contain rows of data with some empty attributes.

This can be a serious problem when we are going to analyze data regarding a particular attribute.

Therefore, a data "cleaning" operation should always be performed initially, especially on the attributes that will be analyzed. This ensures that these attributes contain concrete data in all rows of the dataset.

```
Attributes names:
- movie_id
- movie_name
- year
- certificate
- runtime
- genre
- rating
- description
- director
- director_id
- star
- star_id
- votes
- gross(in $)
```

1. Download from Moodle the folder **FICHA09**, which contains a python file and a sub-folder file with the dataset *films.csv*.
2. Open on VSCode the folder **FICHA09**.
3. Fill in the **loadDataFilms()** function, which should return the list of the movie data, as well as a list of the attributes (the first line of the CSV file).
4. The **showDatasetResume()** function should print a summary of the dataset to be analyzed, as shown below:



```

Number of films : 52452

Dimension of films data:  52452 films

Atributes of dataset:  14 attributes

Attributes names:
- movie_id
- movie_name
- year
- certificate
- runtime
- genre
- rating
- description
- director
- director_id
- star
- star_id
- votes
- gross(in $)
Press any key to continue . . .

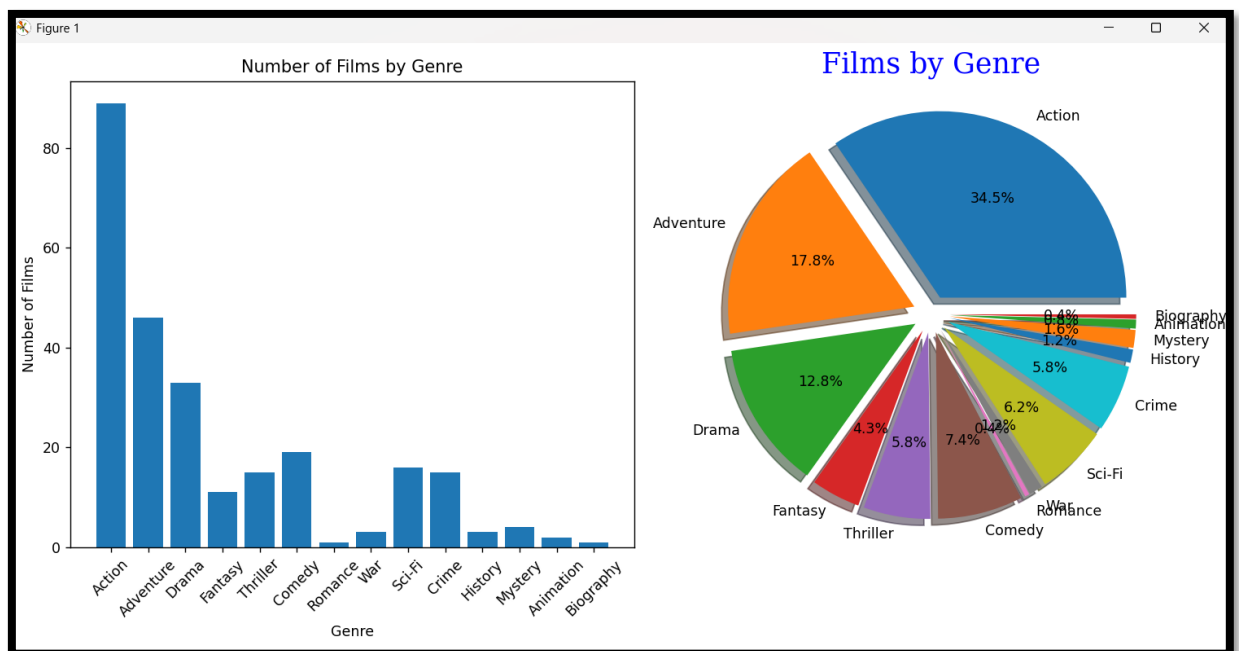
```

5. The **cleanDataset()** function must comply with the following objectives:
 - Since the dataset is very large (52.452 movies), let's limit our list to the **first 100 films of the dataset** (meaning a sample), so that the processing is faster. From now we work only with these records.
 - Considering that in this exercise we will analyze (as an example) the **genre** and **rating** attributes, your function should analyze the first 100 films considered above, and remove all those that do not present valid data in the *genre* or *rating* attributes (namely if any of this attribute is empty, such as "").

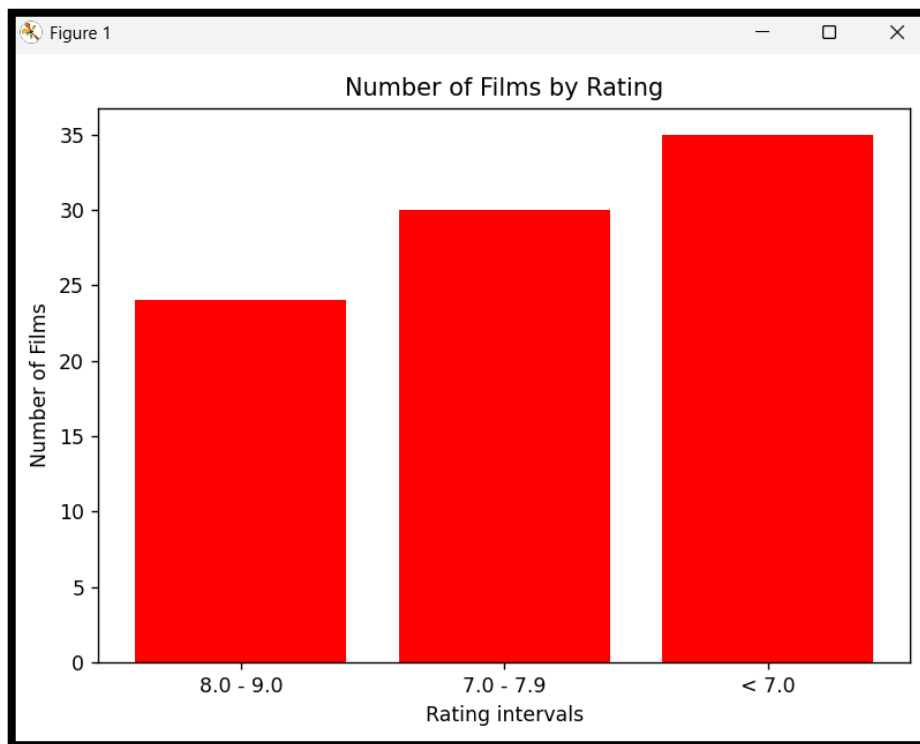
- The function should end with the print of the number of movies after data cleaning, and should **return** the "clean" dataset.

```
Number of films after cleaning: 89
Press any key to continue . . .
```

6. The **showFilmsByGenre()** function should create a figure with two charts (two subplots) that illustrate the number of films per genre.
- The first chart should be a bar chart
 - The second should be a pie chart with the percentage of each genre of film in reference to the total sample
 - Note1: A movie is usually associated with several genres (see CSV file).
 - Note2: the figure is scaled as follows: **plt.figure(figsize=(12,6))**. That means 12 inches width and 6 inches height.



7. The **showFilmsByRating()** function should calculate the number of films in each of the following rating ranges, and illustrated in the chart below:
- Nº of films rated between 8.0 and 9.0
 - Nº of films rated between 7.0 – 7.9
 - Nº of films rated < 7.0



8. The **showTopRating**(*top*) function should display a table with the top of the films with the highest rating.

For example, *showTopRating(10)* should show the top 10 of the highest rated movies, as in the example below:

```

Top 10 films by rating:
- The Lord of the Rings: The Return of the King      Rating: 9.0
- The Dark Knight                                    Rating: 9.0
- Inception                                           Rating: 8.8
- The Lord of the Rings: The Fellowship of the Ring  Rating: 8.8
- The Matrix                                           Rating: 8.7
- Star Wars                                           Rating: 8.6
- Léon: The Professional                             Rating: 8.5
- Gladiator                                           Rating: 8.5
- Avengers: Infinity War                             Rating: 8.4
- Raiders of the Lost Ark                            Rating: 8.4
Press any key to continue . . .

```